Lecture 30

Recap: Learning and Agents

COMP532 – Machine Learning and Bioinspired Optimisation

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What was this course really about?

Core theme

This course studied how intelligent behaviour can emerge from **learning**, **strategic interaction**, **population adaptation**, and **distributed coordination**.

- **Reinforcement learning:** how an agent learns from reward and interaction.
- **Classical game theory:** how rational players reason in strategic settings.
- **Evolutionary game theory:** how behaviours spread in populations over time.
- **Swarm intelligence:** how useful collective behaviour emerges from local rules.

The four paradigms of the course

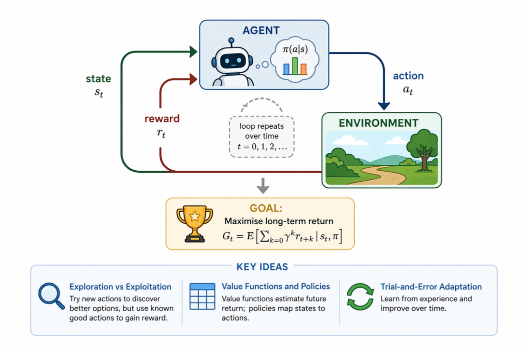
RL	**Classical GT**	**EGT**	**SI**
learning by interaction	strategic reasoning	adaptation in populations	distributed coordination
agent-level update	player-level analysis	population-level dynamics	local interaction + feedback
reward	payoff	fitness	local signal / shared information
sequential decisions	equilibrium analysis	replicator dynamics	emergent behaviour

A unifying comparison

Paradigm	Basic unit	Feedback signal	Adaptation mechanism	Typical question
Reinforcement Learning	agent	reward	value / policy update	What should the agent learn to do?
Classical Game Theory	player	payoff	strategic reasoning	What should rational players choose?
Evolutionary Game Theory	population	fitness	selection / replication	Which behaviours spread over time?
Swarm Intelligence	agent / population	local signals	distributed feedback	How can coordination emerge without central control?

Reinforcement learning in one picture

- An agent interacts with an environment over time.
- At each step, it selects an action and receives a reward.
- The goal is to maximise long-term return.
- Key ideas from the course:
  - exploration vs exploitation
  - value functions and policies
  - trial-and-error adaptation



Classical game theory in one picture

- Multiple players interact strategically.
- Each player's best choice depends on what others do.
- The key object is a **strategy** rather than a policy learned from experience.
- Central concepts:
  - normal-form games
  - best response
  - Nash equilibrium
  - zero-sum versus general-sum interaction

Interpretation

Classical game theory asks what rational players *should* do in strategic settings.

Evolutionary game theory in one picture

- EGT shifts the focus from fully rational players to **populations of interacting types**.
- Strategies with higher fitness become more common over time.
- The central object is the **population state**.
- Key ideas:
  - fitness and selection
  - replicator dynamics
  - evolutionary stability
  - adaptation without explicit deliberation

Interpretation

EGT asks how behaviour changes under repeated interaction and selection pressure.

How are RL, GT, and EGT related?

- **RL** is about *learning through interaction*.
- **Classical GT** is about *strategic reasoning and equilibrium*.
- **EGT** is about *population-level adaptation*.

A useful way to remember the difference

- RL: *How does an agent learn?*
- GT: *What should rational players do?*
- EGT: *Which behaviours survive and spread?*

Where swarm intelligence fits

- Swarm intelligence studies systems with many simple agents.
- Agents use only local information and limited communication.
- There is no central controller.
- Useful global behaviour emerges through:
  - feedback
  - memory
  - stochastic exploration
  - repeated local interaction

Key message

Swarm intelligence is a framework for distributed search and coordination.

Three swarm paradigms from this unit

PSO

- social learning
- continuous search
- shared best + self-memory

ACO

- pheromone trails
- combinatorial search
- environmental memory

Bee-inspired systems

- recruitment
- path integration
- distributed foraging

Choosing the right framework

Use this framework	When the problem looks like this
Reinforcement learning	Sequential decisions, delayed reward, learning from repeated interaction
Classical game theory	Strategic players, equilibrium reasoning, incentive analysis
Evolutionary game theory	Population adaptation, selection pressure, strategy frequencies changing over time
Swarm intelligence	Decentralised search, local communication, robustness and coordination without central control

Common conceptual confusions

- **Reward vs payoff vs fitness**
  - reward: learning signal in RL
  - payoff: utility in a strategic interaction
  - fitness: reproductive or selection advantage in a population
- **Policy vs strategy**
  - policy: action rule used by a learning agent
  - strategy: planned behaviour in a game-theoretic setting
- **Nash equilibrium vs evolutionary stability**
- **Exploration in RL vs exploration in swarm systems**

Exam format and preparation

- **Multiple choice section:** short conceptual questions.
- **Written section:** explanation, comparison, and application.
- Revise not only definitions, but also:
  - differences between paradigms
  - when each method is appropriate
  - core mechanisms and assumptions
  - representative examples

Advice

If you can clearly explain the main concepts, you are in a strong position.

Example multiple-choice question

Question

Which of the following scenarios is **best** modelled using Ant Colony Optimisation rather than Particle Swarm Optimisation?

A) Tuning the continuous gains of a PID controller.
B) Searching for the minimum of a differentiable loss function in $\mathbb{R}^{100}$.
C) Constructing a delivery route that visits a set of cities at minimum cost.
D) Learning a policy through repeated interaction with an unknown environment.

Answer

C

Why?

ACO is most natural for **discrete combinatorial optimisation** problems such as routing and tour construction, whereas PSO is more natural for **continuous black-box optimisation**.

Example written question

Question

Explain the concept of **stigmergy** in swarm-based optimisation.
Illustrate your answer with reference to **Ant Colony Optimisation (ACO)**.

Indicative answer

Stigmergy is a form of indirect coordination in which agents communicate by modifying their shared environment. In **ACO**, ants deposit pheromone on edges, and future ants are biased toward edges with stronger pheromone. In this way, good partial solutions are reinforced over time. Pheromone evaporation provides forgetting, preventing the system from locking too quickly onto poor choices.

Final takeaway

One sentence summary

The course has studied how adaptive behaviour emerges under feedback, interaction, and limited information.

- Reinforcement learning: adaptation by reward.
- Game theory: interaction by strategic reasoning.
- Evolutionary game theory: adaptation by selection.
- Swarm intelligence: coordination by local rules.

Thank you and good luck!